

Adi Nata

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)



PERSONAL SUMMARY

Accomplished software engineer specializing in embedded systems, full-stack development, and IT infrastructure. Demonstrates a proven track record in manufacturing execution systems (MES), high-performance C/C++ libraries, and enterprise-grade web application development. Proficient in cloud technologies, ISO-compliant IT policies, and process optimization.

SKILLS

- Programming Languages: Python, C, C++, C#, JavaScript, Java.
- Web development: React, Svelte, Express, HTML, Tailwind, Spring Boot, .NET.
- Database: PostgreSQL, MySQL, MongoDB, Redis.
- DevOps: Git, Docker, Kubernetes, GitHub Actions.
- Supporting Tools: Ubuntu, AWS, Oracle Cloud, Nginx, CMake, Make, Gtest.
- Core Competencies: Problem Solving, Communication, Collaboration, Adaptability, Mobility.

EXPERIENCES

Senior Research Engineer

*A*Star SIMTech, Jan 2026 - Present*

- Designed and implemented a distributed task orchestration architecture composed of microservices, asynchronous messaging, and API-driven workflow coordination to automate execution across heterogeneous machine systems with .NET Framework.
- Built an event-driven integration backbone using RabbitMQ to decouple React frontend, Python middleware, .NET orchestration, and Python + .NET machine communication layers, enabling scalable and fault-tolerant processing.
- Developed SiLA compliant (gRPC-based) integration services to support standardized machine communication from underlying machine protocol (TCP/IP, serial communication, GUI automated, et cetera).
- Deployed and maintained containerized backend services using Docker with PostgreSQL persistence layers, supporting distributed deployment, background polling services, workflow persistence, and production-grade system reliability.

Software Engineer

Vital Thin Film Materials, Apr 2024 – Dec 2025

- Developed MES web applications using React, Node, and PostgreSQL for process monitoring, achieving an 8% reduction in manufacturing process time and enhancing data reliability.
- Connected MES with FANUC CNC machine data through HTTP pooling and web socket integration with the React frontend.
- Configured network infrastructure, including Meraki router and firewall setup, wireless heatmap creation, and VPN connection planning.
- Implemented a document control and approval system with Microsoft Power Automate, reducing average document approval time by 3 days.
- Authored IT policies such as Disaster Recovery Plan, Incident Response Plan, and Cyber Security Awareness, ensuring compliance with ISO 27701 and Malaysia PDPA standards.

Embedded Software Engineer

Continental Automotive, Feb 2023 - Mar 2024

- Developed a high-performance C and C++ network communication library to enable secure HTTP and MQTT connections between vehicle firmware and central servers, following Agile methodology.
- Implemented unit tests using Google Test, achieving 98% of line coverage and 99% of branch coverage, ensuring code reliability in compliance with ASPICE and MISRA standards
- Automated module flashing and testing using Arduino to simulate physical button presses and toggles on the load box, facilitating overnight testing and reducing average testing time by 10% for common tests.
- Enhanced and automated the firmware build process by optimizing the CMake command and creating bash scripts, reducing compilation and build time from 15 minutes to 5 minutes.

EDUCATION

National University of Singapore (NUS), Computer Engineering

- Minor: Financial Mathematics.
- Activities: Engineering Student Ambassadors.
- CAP: 4.17/5 (Second Class Honors).

St Andrew's Junior College, A-Level

- Activities: Talent Development Program, Student Council.
- Obtained Merit for H3 course at NTU: Semiconductor Physics & Devices (PHYS0001).

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Developer Professional
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional
- Oracle Cloud Infrastructure 2025 Certified Network Professional
- Certified SAFe 6 Practitioner

PROJECTS

Laser Tag

- Multiplayer, multithreaded, handheld laser tag system.
- Tech Stack: Python, C, Arduino.

BestFit

- E-commerce apparel platform with size recommendation feature.
- Tech Stack: React, Express, MongoDB, AWS S3, Stripe, Python sklearn.

ADDITIONAL EXPERIENCES

Research Assistant

National University of Singapore, Jan 2022 - Jun 2022

- Developed a CNN model to detect malaria-infected cells with 91% accuracy and 95% precision.

Business Analyst Intern

Workforce Optimizer Pte Ltd, Jun 2021 - Dec 2021

- Enhanced LSTM forecasting model for manpower allocation, reducing RMSE from 3.4 to 1.2.

Teaching Assistant

National University of Singapore, Jun 2020 – Jun 2022

- Mentored over 100 students in C programming with an average student rating of 4.7/5.

LANGUAGES

- English (Bilingual)
- Indonesian (Native)
- Malay (Native)
- Mandarin (Conversational)